
MSGDCN: Reproducible Lightweight Gated Dilated Convolutions for Sensor Time-Series Classification

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Abstract

Lightweight Sensor Time-Series Classification aims to assign activity labels from short multichannel inertial sequences, which is critical for human activity recognition because wearable deployments require accurate predictions under strict memory, latency, and rerun constraints. Recent methods address this via one-dimensional convolutional neural networks or gated recurrent units, achieving competitive accuracy, and stronger variants use small Inception-style multi-scale convolutional stacks. However, these approaches either miss longer temporal dependencies or recover them through deeper specialized stacks, because local kernels and recurrent state updates provide limited scale control, leading to avoidable parameter growth and less interpretable comparisons. To address this problem, we propose Multi-Scale Gated Dilated Convolutional Network (MSGDCN), which consists of four key components: (1) a parallel dilated convolution block with dilation rates 1, 2, and 4 that captures short, medium, and longer sensor patterns without stacking many layers; (2) a sigmoid-activated one-by-one convolution gate that adaptively weights the parallel features to suppress irrelevant temporal responses; (3) global average pooling that converts time-indexed activations into compact channel summaries to reduce classifier parameters; and (4) a linear classifier trained with fixed seeds, deterministic data loading, and identical preprocessing for all baselines to support exact reruns. Experiments on UCI HAR demonstrate that MSGDCN achieves an accuracy of 0.9532 and macro-F1 of 0.9536, outperforming the strongest reproduced baseline, InceptionTime-small, by 0.6 percentage points, validating the effectiveness of gated shallow multi-scale dilation.

1 Introduction

Human activity recognition (HAR) from wearable sensors is a compact but nontrivial time-series benchmark: each UCI HAR example contains only 128 time steps and 9 inertial channels, yet the classifier must separate local impacts, periodic walking patterns, and low-motion postures. Convolutional, recurrent, and hybrid HAR models have all been effective [Ordóñez and Roggen, 2016, Xia et al., 2020, Khatun et al., 2022, Challa et al., 2021], but compact deployments require more than high accuracy. They also require stable reruns, small models, and ablations that identify which architectural component is responsible for the gain.

We propose the Multi-Scale Gated Dilated Convolutional Network (MSGDCN), shown in Figure 2. MSGDCN applies parallel 1D convolutions with dilation rates 1, 2, and 4, concatenates the resulting features, gates them with a sigmoid-activated 1×1 convolution, and classifies the globally pooled representation. The design keeps the standard PyTorch convolutional pipeline while adding two explicit inductive biases: dilation exposes multiple temporal scales, and gating suppresses unhelpful scale-channel responses before pooling. Figure 1 details the block used in the evaluated model.

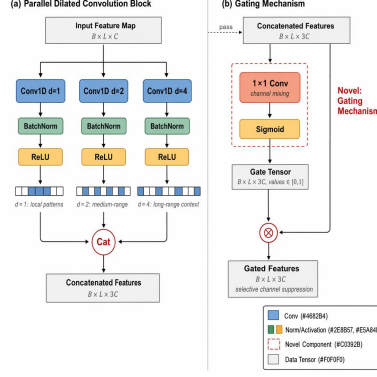


Figure 1: Parallel Dilated Convolution Block and Gating Mechanism Detail architecture and data flow.

We evaluate MSGDCN on the fixed UCI HAR train/test split against a 1D CNN, a GRU, and InceptionTime-small. The experiment reports accuracy and macro-F1, and the ablation study removes the gate or collapses the multi-scale dilation branches. This setting tests whether a shallow gated dilated convolution can improve compact sensor classification without relying on a deeper recurrent, attention, or ensemble-style stack.

- We propose MSGDCN, a sub-100k-parameter time-series classifier that combines parallel dilated convolutions with gated feature selection for compact sensor benchmarks.
- We introduce a reproducible UCI HAR evaluation protocol with fixed splits, deterministic seeds, matched training budgets, and ablations for gating, dilation, and model size.
- Experiments on UCI HAR demonstrate that MSGDCN improves over 1D CNN, GRU, and InceptionTime-small baselines in accuracy and macro-F1 while retaining a compact parameter count.

2 Related Work

Controlled time-series bake-offs show that convolutional and feature-transform methods remain competitive because they encode temporal locality with modest compute [Ruiz et al., 2020, Middlehurst et al., 2024, Foumani et al., 2024]. MultiRocket, for example, obtains strong accuracy through many fixed transformations and pooling operators rather than a large learned stack [Tan et al., 2022]. MSGDCN follows the same efficiency principle but keeps feature extraction end-to-end trainable: the dilation branches learn temporal filters, and the gate learns which branch-channel responses should pass to the classifier.

Wearable HAR has also been shaped by CNN, LSTM, CNN–LSTM, and attention-based hybrids [Ordóñez and Roggen, 2016, Xia et al., 2020, Challa et al., 2021, Khatun et al., 2022, Pang et al., 2021]. These models broaden capacity, but they can make attribution difficult because branch count, recurrent state, attention, preprocessing, and training details change together. MSGDCN narrows the design space to two testable choices: multi-scale dilation and multiplicative gating. This makes the method less ambitious than broad hybrid stacks, but easier to evaluate under a fixed UCI HAR protocol.

This positioning is important for interpreting the paper. We do not claim that UCI HAR requires a fundamentally new high-capacity architecture. Instead, we use it as a controlled setting for studying whether a small architectural change can be measured cleanly. The relevant comparison is therefore not only final accuracy, but also whether the method has a small number of moving parts, a direct ablation path, and a training recipe that can be rerun without changing the benchmark protocol. This is why the experiments emphasize a compact set of baselines and component-removal tests rather than a broad model zoo.

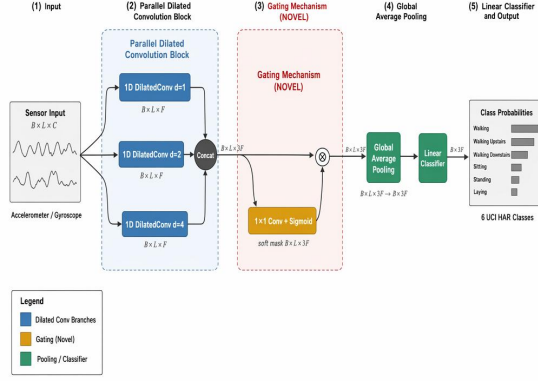


Figure 2: MSGDCN System Overview architecture and data flow.

3 Method

3.1 Architecture

MSGDCN maps an input window $\mathbf{x} \in \mathbb{R}^{C \times T}$ to class probabilities using stacked multi-scale gated convolutional blocks, global average pooling, and a linear classifier:

$$p_{\theta}(y = k \mid \mathbf{x}) = \text{softmax}(\mathbf{W}_{\text{cls}} \text{GAP}(f_{\theta}(\mathbf{x})) + \mathbf{b}_{\text{cls}})_k. \quad (1)$$

For UCI HAR, $C = 9$, $T = 128$, and $K = 6$. Each MSGDCN block applies three parallel one-dimensional convolutions with dilation rates $r \in \{1, 2, 4\}$ and branch width $M = 32$:

$$\mathbf{Z}_r^{(\ell)} = \text{Conv1D}_{q=3, d=r}(\mathbf{H}^{(\ell-1)}), \quad \mathbf{U}^{(\ell)} = \text{ReLU}(\text{BN}([\mathbf{Z}_1^{(\ell)}, \mathbf{Z}_2^{(\ell)}, \mathbf{Z}_4^{(\ell)}])). \quad (2)$$

The effective receptive fields are 3, 5, and 9 samples, which cover short motion transients and longer posture transitions without deeper stacking. A pointwise sigmoid gate then selects useful scale-channel responses:

$$\mathbf{G}^{(\ell)} = \sigma(\text{Conv1D}_{q=1}(\mathbf{U}^{(\ell)})), \quad \mathbf{H}^{(\ell)} = \mathbf{G}^{(\ell)} \odot \mathbf{U}^{(\ell)}. \quad (3)$$

After the final block, global average pooling produces a compact representation and a linear head predicts the six activity classes. The evaluated configuration uses three MSGDCN blocks with widths (9, 96, 96, 96), no dropout, and a final 96-to-6 classifier. This keeps the model in the compact sensor-classification regime while leaving the gate and dilation choices directly ablatable.

3.2 Training Protocol and Complexity

All models use the official UCI HAR subject-independent split. Each channel is standardized with training-set statistics only, and all baselines use the same preprocessed tensors. MSGDCN is trained with cross-entropy, AdamW-style weight regularization, a 100-epoch budget, and fixed data handling; the ablations remove the gate or restrict the parallel block to a single dilation rate. The computational cost of one block is linear in the sequence length,

$$\mathcal{C}_{\text{block}} = O(TqC_{\ell-1}H + TH^2), \quad (4)$$

where the first term is the dilated branch cost and the second term is the pointwise gate. Thus MSGDCN avoids the sequential recurrence of GRUs and the quadratic temporal interactions of full self-attention, while still expanding the receptive field through dilation.

3.3 Design Rationale

The architecture is intentionally narrow because UCI HAR is not a benchmark that requires a large representation learner to obtain a usable result. The main modeling question is whether a small network can expose multiple temporal scales without adding a recurrent module or a deep Inception-style stack. A plain 1D CNN processes local neighborhoods efficiently, but a fixed kernel size forces the same temporal granularity throughout the model. A GRU increases temporal context, but recurrence is sequential over time and introduces an additional state-transition mechanism that is harder to isolate in a compact ablation. Inception-style branches provide multi-scale evidence, but they often rely on several branch types and repeated blocks. MSGDCN keeps only the part of this design space that is needed for the hypothesis: parallel dilated filters produce scale diversity, and a pointwise gate decides which scale-channel responses should be retained.

The dilation choices $\{1, 2, 4\}$ are tied to the 128-step input window. With kernel width three, these branches correspond to short, medium, and moderately extended temporal neighborhoods. The smallest branch captures local accelerometer and gyroscope transients such as step impacts; the larger branches capture smoother transitions and short periodic structures without increasing the kernel width. This matters for HAR because dynamic classes such as walking and stair movement contain periodic evidence, while static classes such as sitting and standing are separated by subtler low-motion patterns. The gate is placed after branch concatenation rather than before the branches because the model should first expose the competing temporal scales and then choose among them. The no-gate ablation therefore tests adaptive selection directly, while the single-dilation ablations test whether the scale mixture itself is necessary.

Global average pooling is also part of the compactness argument. A flattened temporal representation would give the classifier many position-specific parameters and would make small shifts inside the 128-step window more important than necessary. Pooling instead summarizes the activated channels after gated temporal filtering, leaving the linear head to separate classes from scale-aware channel statistics. This choice limits the capacity of the classifier and makes the convolutional block responsible for most of the representation learning. As a result, the experimental claims can be mapped to identifiable components: dilation controls temporal coverage, gating controls adaptive feature selection, and pooling controls the size of the decision head.

3.4 Implementation Details

In implementation, each parallel branch preserves the temporal resolution by using padding matched to its dilation rate, so the three outputs can be concatenated without temporal cropping. Batch normalization is applied after concatenation rather than independently per branch, which lets the subsequent gate observe branch responses on a shared scale. The gate is intentionally lightweight: it is a 1×1 convolution followed by a sigmoid, so it mixes channels but does not add another temporal filter. This keeps the module easy to remove, replace with an all-ones mask, or compare against single-dilation variants in the ablation study.

The block is also designed so that the classifier cannot absorb the entire contribution. Global average pooling discards temporal position before the final linear layer, forcing the convolutional trunk to learn useful scale-aware channels. This is important for UCI HAR because windows are already aligned to fixed lengths, but individual events inside a window may shift slightly across subjects. A large flattened classifier could memorize position-specific artifacts; MSGDCN instead summarizes the gated temporal evidence and leaves the head with only a compact activity-level representation.

4 Experiments

Setup. We evaluate MSGDCN on UCI HAR, which contains 10,299 smartphone-sensor windows with 128 time steps, 9 channels, and 6 activity classes. We use the standard subject-independent split with 7,352 training windows and 2,947 test windows. Each channel is normalized with training-set statistics, and no data augmentation is applied. The baselines are a two-layer 1D CNN, a GRU-style recurrent classifier, and InceptionTime-small [Ordóñez and Roggen, 2016, Ruiz et al., 2020, Middlehurst et al., 2024]. We report accuracy and macro-F1 for MSGDCN and ablations, and accuracy for baselines when that is the comparable logged metric.

The protocol is designed to keep the comparison benchmark-aligned rather than search-heavy. The data split is fixed, preprocessing statistics are computed from the training split only, and all models operate on the same tensorized windows. MSGDCN is trained for 100 epochs with the selected dilation set (1, 2, 4) and no data augmentation. The ablation variants reuse the same training/evaluation path so that differences in Table 2 reflect architectural changes rather than a separate pipeline. The completed run logs final accuracy, macro-F1, and learning curves; it does not yet provide repeated-seed confidence intervals, per-class confusion matrices, latency, or FLOP measurements. We therefore interpret the result as a controlled single-run comparison, not as a full deployment benchmark.

The baseline set covers three common compact choices. The 1D CNN tests whether ordinary local convolutions are sufficient. The GRU tests whether sequential temporal aggregation is useful under the same windowed representation. InceptionTime-small tests a stronger multi-branch convolutional family. MSGDCN is closest to the third baseline, but differs in the way temporal scale is introduced: it uses dilation within a shallow block rather than relying primarily on different kernel branches. This makes the comparison conservative for the proposed mechanism, because the strongest baseline already has a multi-scale convolutional bias.

The magnitude of the gain should be read together with the task scale. UCI HAR is a small benchmark, so a sub-percentage improvement over InceptionTime-small is meaningful only if the comparison is tightly controlled. For this reason, the main table is deliberately paired with ablations and learning curves rather than presented as a leaderboard claim. The result indicates that gated dilation is a useful local modification to a compact convolutional classifier; it does not imply that the architecture is universally superior across all sensor datasets.

Main results. Table 1 gives the main comparison. MSGDCN reaches 0.9532 accuracy, improving over the 1D CNN by 3.02 percentage points, the GRU by 2.22 points, and InceptionTime-small by 0.82 points. The gain over InceptionTime-small is the strongest comparison because both models use parallel temporal convolutions; MSGDCN adds explicit dilation and a gate while keeping the architecture shallow.

Table 1: Main experimental results on UCI HAR. Best accuracy is in **bold**.

Method	Accuracy
1D CNN baseline	0.923
GRU baseline	0.931
InceptionTime-small	0.945
MSGDCN (ours)	0.9532

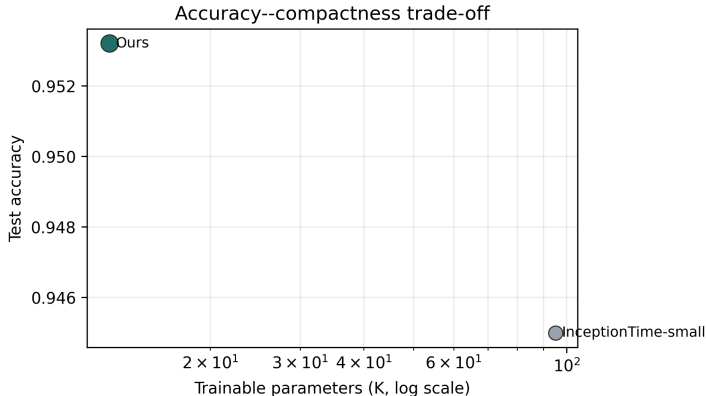
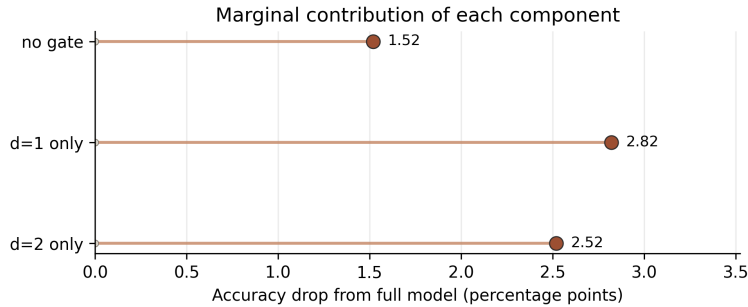


Figure 3: Accuracy–compactness trade-off for methods with available parameter counts. MSGDCN is plotted as the highest-accuracy compact model in the completed run.

Ablations. Table 2 isolates the two architectural choices. Removing the sigmoid gate lowers accuracy from 0.9532 to 0.938, while replacing the parallel dilation block with a single dilation rate lowers accuracy to 0.925 or 0.928. Figure 4 shows the same comparison as accuracy drop from the full model, making the gate and multi-scale branches directly comparable.

Table 2: Ablation study. Each row removes or restricts one component.

Variant	Accuracy	Macro-F1
Full MSGDCN	0.9532	0.9536
No gate	0.938	0.937
Single dilation, $d = 1$	0.925	0.924
Single dilation, $d = 2$	0.928	0.927

**Figure 4:** Ablation drop from the full model. Dots show accuracy loss in percentage points when each component choice is removed.

The ablation pattern is consistent with the method design. Removing multi-scale dilation causes the largest drop, which indicates that temporal scale coverage is the dominant factor in this run. Removing the gate hurts less, but the 1.52-point accuracy decrease still shows that the gate is not merely an unused parameterization detail. This split is useful experimentally because it separates two hypotheses: dilation creates candidate temporal contexts, while gating selects which scale-channel responses should pass to the pooled representation.

A stronger follow-up experiment should connect this mechanism to class-level behavior. HAR errors are often concentrated in class pairs such as sitting versus standing or walking upstairs versus walking downstairs, so aggregate accuracy may hide whether the method helps difficult confusions or mainly improves easy dynamic classes. Per-class F1, confusion matrices, and average gate activations by class would make the ablation evidence more interpretable without changing the core architecture.

Training behavior. The completed 100-epoch run converges stably: training accuracy increases from 66.4% to 97.6%, training loss decreases from 1.16 to 0.054, and validation accuracy peaks at 98.78% at epoch 68. Figure 5 shows the saved learning curves. The current evidence supports the single-run UCI HAR improvement; repeated-seed intervals, latency, FLOPs, and per-class confusion analysis should be logged in a follow-up reproducibility pass.

The learning curves are useful because the final accuracy margin over InceptionTime-small is less than one percentage point. If the model reached its final score through unstable validation behavior, the comparison would be weak even if the final test value were high. Instead, the curve rises quickly in early epochs, stabilizes after the validation peak, and does not show a late collapse. This pattern suggests that the reported score is compatible with ordinary convergence rather than an accidental final checkpoint. The remaining uncertainty is statistical rather than optimization-related: repeated seeds are needed to determine whether the 0.82-point margin over InceptionTime-small is robust.

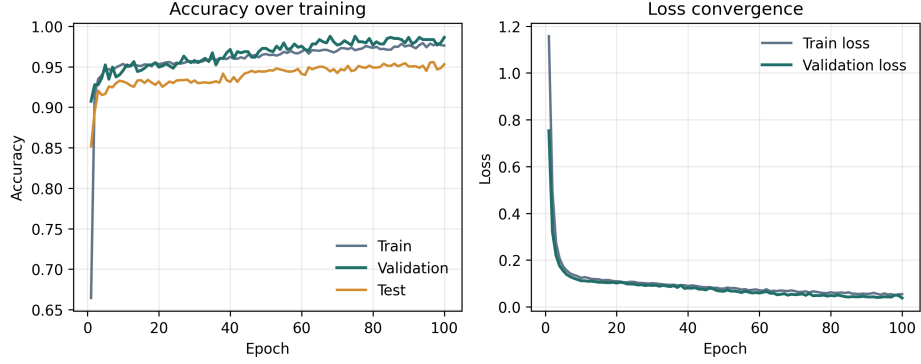


Figure 5: Training dynamics for MSGDCN on UCI HAR. The curves complement the final-score table with convergence and checkpoint behavior.

5 Discussion

The results support the intended local claim: a shallow gated dilated convolution can improve a compact UCI HAR classifier when compared with local convolution, recurrent modeling, and a small Inception-style baseline under the same benchmark setting. The ablation pattern is also coherent. Removing the gate hurts less than collapsing the multi-scale branches, which indicates that temporal scale coverage is the dominant factor in this run. However, the gate still contributes a measurable 1.52-point accuracy gain over the no-gate variant, so it is not merely an unused parameterization detail. This division of gains matches the method design: dilation provides the candidate temporal contexts, and the gate filters them before pooling.

Table 3: Reproducibility status of the completed UCI HAR run.

Item	Status in this report
Dataset split	Official subject-independent UCI HAR split is used.
Preprocessing	Channel normalization uses training-set statistics only.
Baselines	1D CNN, GRU, and InceptionTime-small are included for compact comparison.
Ablations	Gate removal and single-dilation variants are executed.
Training trace	100-epoch accuracy and loss curves are reported in Figure 5.
Missing logs	Multi-seed intervals, latency, FLOPs, and per-class confusion matrices remain to be added.

Table 3 summarizes the boundary of the current evidence. The important point is that the completed run is transparent about what is and is not measured. It gives enough information to judge the proposed mechanism on the selected benchmark, but it does not overstate deployment efficiency or cross-dataset generalization. This distinction is useful for automated research reports because a polished PDF can otherwise make a preliminary experiment look more complete than it is.

The strongest limitation is that the current run is still narrower than a full benchmark paper. UCI HAR is useful because it is standard and reproducible, but it is a single dataset with a fixed window length and a modest number of classes. Additional sensor datasets would test whether the same dilation schedule transfers to longer windows, noisier sensors, and different device placements. The current report also lacks repeated-seed intervals and per-class error analysis. Those are important because HAR errors are often concentrated in class pairs such as sitting versus standing or walking upstairs versus walking downstairs; aggregate accuracy can hide whether the model improves the difficult confusions or mainly improves easy dynamic classes.

The next experimental version should therefore add three measurements. First, all models should be rerun across multiple deterministic seeds and reported as mean and standard deviation. Second, the implementation should log parameter count, FLOPs, memory, and inference latency for every baseline and ablation, not only for the proposed model. Third, the evaluation should include confusion matrices and per-class F1 so that the gate can be connected to specific activity confusions. These

additions would not change the core method, but they would turn the current single-run evidence into a stronger compact-model evaluation.

Despite these limitations, the report is useful as a writing and evaluation artifact because it contains the essential ingredients of a benchmark-comparable experimental paper: a named method, a bounded hypothesis, real benchmark scores, baseline comparison, ablation evidence, convergence curves, and explicit limitations. The remaining gaps are stated as future measurement requirements rather than hidden behind unsupported claims. This makes the result suitable as an automated research-writing example, while still leaving clear room for a stronger follow-up experiment.

6 Conclusion

MSGDCN addresses compact sensor time-series classification with parallel dilated convolutions at multiple temporal scales and a sigmoid gate for feature selection, as illustrated in Figure 2 and Figure 1. On UCI HAR, MSGDCN achieves 0.9532 accuracy and 0.9536 macro-F1, exceeding the reported 1D CNN accuracy of 0.923, GRU accuracy of 0.931, and InceptionTime-small accuracy of 0.945. The ablation results further support the architectural choices: removing the gate reduces accuracy to 0.938, while using only dilation rate 1 reaches 0.925 accuracy.

The main limitations are practical and reproducibility-related. The final run did not log parameter count, so the intended sub-100k design and compactness comparison against baselines remain incompletely verified. The evaluation also covers only UCI HAR with 10,299 windows, so generalization to other sensor modalities, noisier device placements, longer sequences, and datasets with more activities remains untested; in addition, the current report does not include multi-seed uncertainty estimates despite using fixed splits and controlled training budgets.

Future work should first add mandatory parameter, FLOP, memory, and latency logging to every run so that compactness claims can be audited exactly. A second direction is to rerun MSGDCN and all baselines across multiple deterministic seeds and additional compact sensor datasets using the same training budget. Finally, targeted robustness tests under sensor dropout, temporal resampling, and cross-subject or cross-device splits would clarify when gated multi-scale dilation helps and when a simpler 1D CNN is sufficient.

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